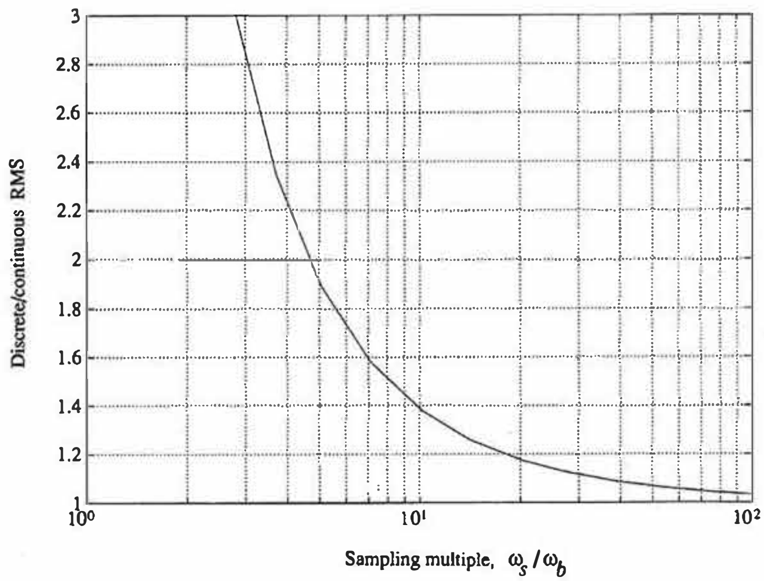


Figure 11.5
Discrete controller degradation versus sample rate for full state feedback and driven by a white disturbance, Example 11.3

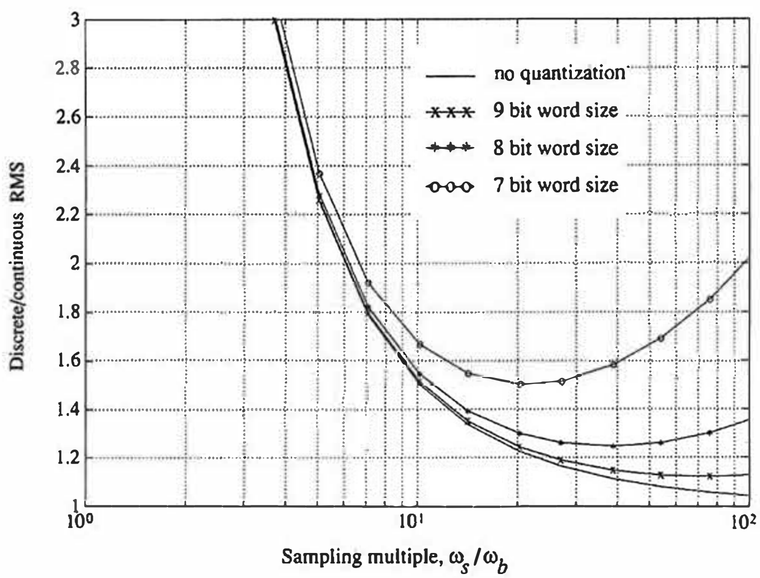


Figure 11.6
Discrete controller degradation versus sample rate for the case with an estimator with quantization errors and a white disturbance, Example 11.3.

curve in Fig. 11.6 shows the discrete to continuous rms ratio and, compared to Fig. 11.5, is slightly higher. This is to be expected because the velocity information is now being estimated from a position measurement and the greater sensitivity to disturbances is the result of the approximate differentiation that is occurring. Although the curve is generally higher at all sample rates, the conclusion concerning sample rate selection for the case where disturbances are the dominant consideration is the same; that is, sample at 20 times the bandwidth or higher.

quantization

The addition of the random noise from quantization shows that there are limits to the improving noise response as the sampling rate increases. Figure 11.6 also includes quantization noise added according to the discussion in Section 10.1. Quantization is usually important only with a fixed-point implementation of the control equations, and an assumption was therefore required as to the scaling of the signals for this analysis. In Fig. 11.6 it is assumed that the controller was scaled so that the continuous rms error due to the plant disturbance is 2% of full scale, a somewhat arbitrary assumption. But the point is not the specific magnitude as much as the notion that there is a limit, and that if a designer is dealing with a microprocessor with fewer than 12 bits, it can be useful to perform a similar analysis to determine whether the word-size errors are sufficiently large to impact selection of sample rate. With 16- and 32-bit microprocessors and a parallel or cascade realization, the increase in rms errors due to quantization at the fast sample rates is typically so small that word size is not an issue and no practical upper limit to the sample rate exists. However, if using an 8-bit microprocessor, it may be counterproductive to use too high a sample rate.

◆

The example, although on a very simple plant, shows the basic trend that usually results when a white disturbance acts on the plant: The degradation due to the discrete nature of the control over that possible with a continuous control is significant when sampling slower than 10 times the bandwidth. Except for controllers with small word sizes (8 bits or less), the performance continues to improve as the sample rate is increased, although diminishing returns tend to occur for sampling faster than 40 times the bandwidth.

Whether the rms errors due to plant disturbances are the primary criterion for selecting the sample rate is another matter. If cost was of primary importance and the errors in the system from all sources were acceptable with a sample rate at three times bandwidth, nothing encountered so far in this chapter would necessarily prevent the selection of such a slow sample rate.

resonances

On the other hand, resonances in the plant that are faster than the bandwidth sometimes can have a major impact on sample rate selection. Although they do not change the fundamental limit discussed in Section 11.1, they can introduce unacceptable sensitivities to plant disturbances. The analysis is identical to that used in the example above, except that it is mandatory in this case that an accurate evaluation of the integral in Eq. (11.7) be used because there are some plant dynamics that are faster than the sample period.

◆ **Example 11.4** *Double Mass-Spring Disturbance Response vs. Sample Rate*

Repeat Example 11.3 for the double mass–spring system that was used in Examples 8.3 and 9.5 and described in Appendix A.4. Do part (a) for full-state feedback and part (b) for an estimator based on a measurement of d, that is, the noncolocated case where the resonance is between the sensor and the actuator.

Solution. The parameter values used are as given in Example 8.3, except that we will consider the case with no natural damping as well as the lightly damped ($\zeta = 0.02$) case used previously. The optimal continuous design was carried out in order to achieve a 6:1 ratio between the resonant mode frequency, ω_r, and the system bandwidth, ω_b. Weighting factors were applied to d and y, which provided good damping of the rigid body mode and increased slightly the damping of the resonance mode. The family of discrete designs were found by using pole placement so that the discrete roots were related to the continuous roots by $z = e^{sT}$. The result for the full state feedback case is shown in Fig. 11.7. The general shape of the curves is amazingly similar to the previous example, but there are three highly sensitive sample rates at $2\,\omega_r$, ω_r, and $\omega_r/2$. Note that these are *not* resonances in the traditional sense. The curves represent the ratio of the response of the discrete system compared to the continuous, so that large spikes mean that the discrete controller at that sample rate exhibits poor disturbance rejection compared to the continuous controller. The excitation is broad band for both controllers at all sample rates. Also note that there is *no* peak at $2\,\omega_r$ for the case with $\zeta = 0.02$.

The origin of the sensitivity peak at exactly ω_r is that the controller has no information about the resonance at these critical sample rates. The sampled resonance value will be constant

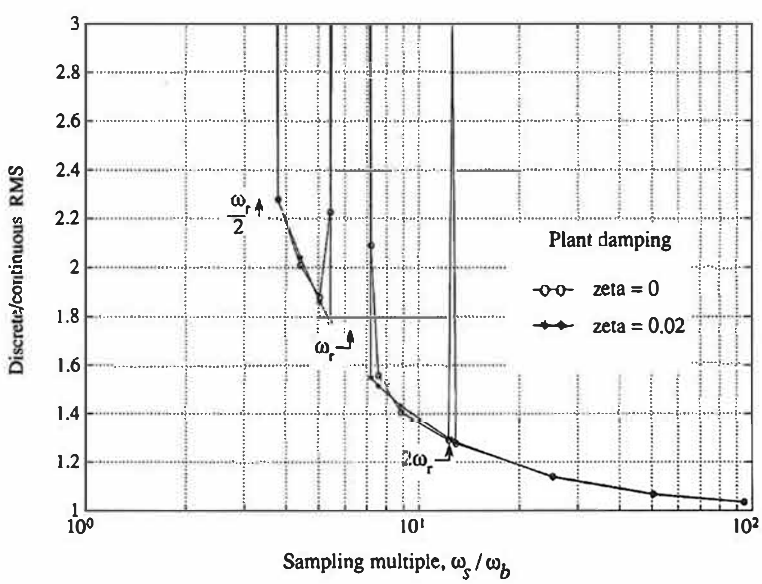


Figure 11.7
Discrete controller degradation for the double mass–spring system using full state feedback, Example 11.4

no matter what the phase relationship between sampler and the resonant mode, thus producing an unobservable system.[3] The additional peak at $2\omega_r$ arises from a similar unobservability, except that in this case the unobservability occurs only when the phasing is such that the samples are at the zero crossings. This peak is significantly less likely to occur and vanishes with a small amount of natural plant damping.

Figure 11.8 shows the sensitivity of the discrete controller to disturbance noise for the more realistic case where the full state feedback has been replaced with an estimator using the measurement of d (essentially a notch filter). As might be expected, the sensitivity peaks have become more serious. The figure shows a wider sample rate band of high sensitivity about each of the previous trouble points and additional peaks at $2\omega_r$ and $\frac{2}{3}\omega_r$. So we now see that there are sensitive sample rates at all integer fractions of $2\omega_r$ due to both unobservabilities noted above.

◆

Example 11.4 demonstrates that, in some cases, a digital controller with sample rates centered about $2\omega_r$ or integer fractions of that value has difficulty and amplifies the effect of random disturbances acting on the plant. Factors that influence whether these sensitive sample rates exist have been studied by Hirata (1989) and generally show that the amount of damping added by the controller is the key aspect. Had more damping been added to the resonant modes by the

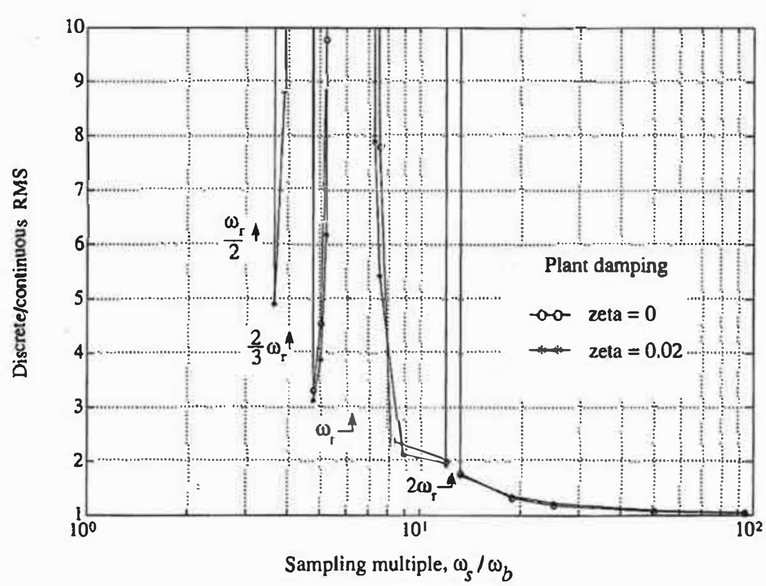


Figure 11.8
Discrete controller degradation for the double mass–spring system using an estimator with the measurement from d, Example 11.4

3 The phenomenon was noted and discussed by Katz (1974) and analyzed by Hirata (1989).

controller than shown by the example, the sensitivity peaks would have been more pronounced because use of the mode information by the controllers would have been more critical to the performance. Furthermore, use of a colocated sensor and actuator—that is, using the measurement of y instead of d—produces results that are very little different than those shown for the noncolocated case, provided that the controller is adding some damping to the resonant mode.

The impact of this is that, for systems where the controller is adding some damping to a lightly damped mode, the only safe place to select a sample rate is faster than twice the resonant frequency, that is

$$\omega_s > 2\omega_r. \tag{11.10}$$

There is a possibility to pick the sample rate in one of the "valleys" in Fig. 11.8; however, resonant frequencies can often vary considerably, thus rendering this approach unreliable, and would cause this system to lack "robustness." For systems with high-frequency resonant modes with adequate natural damping and to which the controller adds no damping, there are typically no sensitive sample rates related to the resonant mode, and these considerations can be ignored when selecting the sample rate. Note that these resonant modes may be considerably faster than the system bandwidth and, therefore, sampling faster than 20 times bandwidth may be advisable in order to eliminate the possibility of these sensitive sample rates.

In summary, the sample rate has a major impact on how well a digital controller performs in the presence of plant disturbances. Performance approaching that of a continuous controller can be achieved providing the sample rate is 20 or more times faster than the bandwidth for systems with no significant dynamics faster than the bandwidth. For systems where the controller is adding some damping to a lightly damped mode that is faster than the bandwidth, the sample rate should also be at least twice that resonant frequency.

11.4 Sensitivity to Parameter Variations

Any control design relies to some extent on a knowledge of the parameters representing plant dynamics. Discrete systems generally exhibit an increasing sensitivity to parameter errors for a decreasing ω_s. The determination of the degree of sensitivity as a function of sample rate can be carried out by accounting for the fact that the model of the plant in the estimator (or compensation) is different from the actual plant.

In Section 8.3.1, we assumed that the actual plant and the model of the plant used in the estimator were precisely the same. This resulted in the separation principle, by which we found that the control and estimation roots designed independently remained unchanged when the feedback was based on the estimated state. We will now proceed to revise that analysis to allow for the case where the parameters representing the plant model in the estimator are different from those of the actual plant.

Let us suppose the plant is described as

$$\mathbf{x}(k+1) = \mathbf{\Phi}_p\mathbf{x}(k) + \mathbf{\Gamma}_p\mathbf{u}(k),$$
$$\mathbf{y}(k) = \mathbf{H}\mathbf{x}(k), \tag{11.11}$$

and the current estimator and controller as

$$\hat{\mathbf{x}}(k) = \bar{\mathbf{x}}(k) + \mathbf{L}(\mathbf{y}(k) - \mathbf{H}\bar{\mathbf{x}}(k)),$$
$$\bar{\mathbf{x}}(k+1) = \mathbf{\Phi}_e\hat{\mathbf{x}}(k) + \mathbf{\Gamma}_e\mathbf{u}(k),$$
$$\mathbf{u}(k) = -\mathbf{K}\hat{\mathbf{x}}(k). \tag{11.12}$$

In the ideal case, $\mathbf{\Phi}_p = \mathbf{\Phi}_e\ (= \mathbf{\Phi})$ and $\mathbf{\Gamma}_p = \mathbf{\Gamma}_e\ (= \mathbf{\Gamma})$, and the system closed-loop roots are given by the controller and the estimator roots designed separately; that is, they are the roots of the characteristic equation [similar to Eq. (8.55)]

$$|z\mathbf{I} - [\mathbf{\Phi} - \mathbf{\Gamma}\mathbf{K}]\,||\,z\mathbf{I} - (\mathbf{\Phi} - \mathbf{L}\mathbf{H}\mathbf{\Phi})| = 0. \tag{11.13}$$

If we allow $\mathbf{\Phi}_e \neq \mathbf{\Phi}_p$ and $\mathbf{\Gamma}_e \neq \mathbf{\Gamma}_p$, the roots do not separate, and the system characteristic equation is obtained from the full $2n \times 2n$ determinant that results from Eqs. (11.11) and (11.12)

$$\begin{vmatrix} \mathbf{\Phi}_p - \mathbf{I}z & -\mathbf{\Gamma}_p\mathbf{K} \\ \mathbf{L}\mathbf{H}\mathbf{\Phi}_p & (\mathbf{I} - \mathbf{L}\mathbf{H})(\mathbf{\Phi}_e - \mathbf{\Gamma}_e\mathbf{K}) - \mathbf{L}\mathbf{H}\mathbf{\Gamma}_p\mathbf{K} - \mathbf{I}z \end{vmatrix}. \tag{11.14}$$

Either the pole-placement approach of Chapter 8 or the steady-state, optimal, discrete design method of Chapter 9 could be used to arrive at the **K**- and **L**-matrices. In both cases, the root sensitivity is obtained by assuming some error in $\mathbf{\Phi}$ or $\mathbf{\Gamma}$ (thus $\mathbf{\Phi}_e \neq \mathbf{\Phi}_p$ and $\mathbf{\Gamma}_e \neq \mathbf{\Gamma}_p$) and comparing the resulting roots from Eq. (11.14) with the ideal case of Eq. (11.13).

If a system has been designed using the methods of Chapter 7, root sensitivity is obtained by repeating the closed-loop root analysis with a perturbed plant or, if one parameter is particularly troublesome, an analysis of a root locus versus that parameter might be worthwhile.

◆ **Example 11.5** *Robustness vs. Sample Rate*

The equations of pitch motion of a high-performance aircraft where there is some fuselage bending are[4]

$$\dot{x}_1 = M_q q + M_\alpha x_2 + M_{\delta_e}\delta_e,$$
$$\dot{x}_2 = q + Z_\alpha + Z_{\delta_e}\delta_e,$$
$$\dot{x}_3 = \omega_r x_4,$$
$$\dot{x}_4 = -\omega_r x_3 - 2\zeta_r\omega_r x_4 + \omega_r K_1 Z_{\delta_e}\delta e + K_2\omega_r Z_\alpha \dot{x}_2, \tag{11.15}$$

4 This example is due to Katz (1974).

where

$$\begin{aligned}
x_1 &= \text{pitch rate},\\
x_2 &= \text{angle of attack},\\
x_3, x_4 &= \text{position and velocity of bending mode},\\
\delta_e &= \text{elevator control surface},\\
M's, Z's &= \text{aircraft stability derivatives},\\
\zeta_r &= \text{bending-mode damping},\\
\omega_r &= \text{bending-mode frequency, and}\\
K_1, K_2 &= \text{quantities depending on the specific aircraft shape}\\
&\quad\ \text{and mass distribution.}
\end{aligned}$$

In the flight condition chosen for analysis (zero altitude, Mach 1.2), the open-loop rigid-body poles are located at $s = -2 \pm j13.5$ rad/sec, and the bending mode is lightly damped ($\zeta_r = 0.01$) with a natural frequency of 25 rad/sec (4 Hz). The control system consists of a sensor for x_1 (a rate gyro), an estimator to reconstruct the remainder of the state, and a feedback to the elevator to be added to the pilot input. The purpose is to change the natural characteristics of the aircraft so that it is easier to fly, sometimes referred to as **stability augmentation**.

The closed-loop poles of the rigid body were located by optimal discrete synthesis to $s = -16 \pm j10$ rad/sec with essentially no change in the bending-mode root locations. The optimal compensator (control law plus estimator) generates a very deep and narrow notch filter that filters out the unwanted bending frequencies. The width of the notch filter is directly related to the low damping of the bending mode and the noise properties of the system. Furthermore, the optimal estimator gains for the bending mode are very low, causing low damping in this estimation error mode. If the bending frequency of the vehicle varies from the assumed frequency, the components of the incoming signal to the estimator due to the bending miss the notch and are transmitted as a positive feedback to the elevator.

Examine the sensitivity to errors in the bending frequency versus the sample rate.

Solution. Figure 11.9 shows the closed-loop bending mode damping ζ_r as a function of the ratio of the sample rate ω_s to bending mode frequency ω_r. Note the insensitivity to sample rate when knowledge of the bending mode is perfect (no error in ω_r) and the strong influence of sample rate for the case where the bending mode has a 10% error. For this example, the bending mode was unstable ($\zeta_r < 0$) for all sample rates with a 10% ω_r error, indicating a very sensitive system and totally unsatisfactory design if ω_r is subject to some error or change from flight conditions.

robust design

The sensitivity of the controlled system to errors in ω_r can be reduced using any one of many methods for sensitivity reduction, which are often referred to as **robust design** methods. One particularly simple method entails increasing the weighting term in $\mathbf{Q}_1$ which applies to the bending mode. The effect of this approach is to increase the width of the notch in the controller frequency response, thus making the controller more tolerant to changes in the bending-mode frequency. Figure 11.10 shows the effect on ζ_r as a function of the fourth element in $\mathbf{Q}_1$, which we will refer to as the bending-mode weighting factor $Q_{1,44}$. It demonstrates a substantial increase in robustness. Note that it is theoretically possible to have $\zeta_r > 0.05$ with a 10% ω_r error and a sample rate only 20% faster than the bending mode; however, this ignores the disturbance-noise amplification that will occur at $\omega_s \cong 2\,\omega_r$ discussed in the previous section.

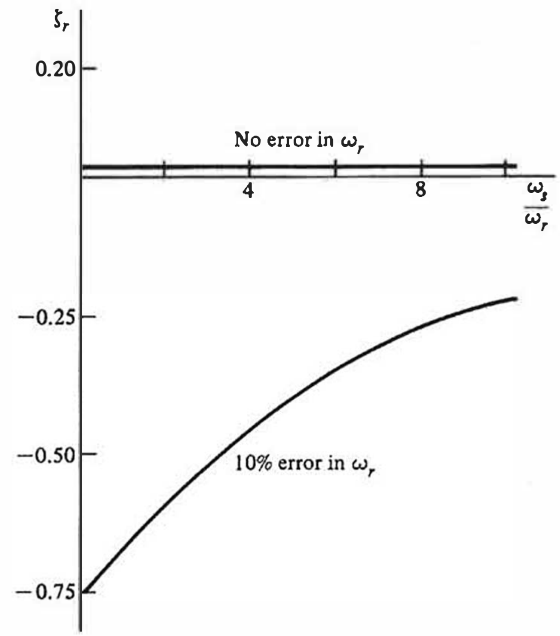


Figure 11.9 Closed-loop bending-mode damping of Example 11.5 versus ω_s

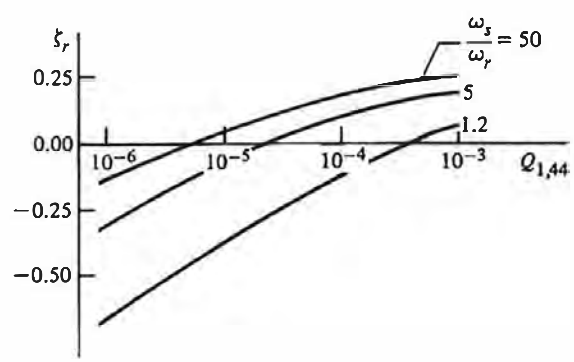


Figure 11.10 Closed-loop bending-mode damping versus $Q_{1,44}$ for Example 11.5 with a 10% ω_r error

With a sufficiently high value of $Q_{1,44}$, the example becomes "robust" in the sense that it is stable for a 10% ω_r error, but it continues to show an increasing sensitivity to the ω_r error for decreasing sample rate.

In summary, for the ideal case where the plant parameters are known exactly, there is no effect of sample rate on bending-mode damping or any other closed-loop dynamic characteristic. On the other hand, if there is some error between the plant parameters used for the controller design and the actual plant parameters, there will be an error in the desired closed-loop characteristics that increases with the sample period. In most cases, the use of reduced performance requirements or of robust design practice such as shown by the example can reduce the error to acceptable levels and thus does not impose extra criteria on the sample rate. However, sensitivity of the system to off-nominal parameters should be evaluated, and in some cases it might be necessary to design specifically for better robustness and, in rare cases, to increase the sample rate over that suggested by other considerations.

11.5 Measurement Noise and Antialiasing Filters

In addition to the random plant disturbances, **w**, that were evaluated in Section 11.3, there are usually some errors in the measurement or unmodeled bending mode oscillations as indicated by **v** in Eq. (9.82). In this section, we wish to examine the effect of the sample rate on the response of the system to measurement errors. These errors are affected significantly by the presence of analog filters (called **antialiasing filters** or **prefilters**), which are typically placed between an analog sensor and the sampler in order to reduce aliasing of high-frequency components of the signal. Therefore, we will discuss antialiasing filters as well.

prefilters

Figure 11.11 depicts the typical arrangement of the antialiasing filter. Digital control systems are arranged this way for many cases in order to prevent the aliasing of the higher-frequency components of **v**. Exceptions are those cases where the sensor is fundamentally digital so that there is no analog signal that can contain high frequency noise; for example, an optical encoder provides a

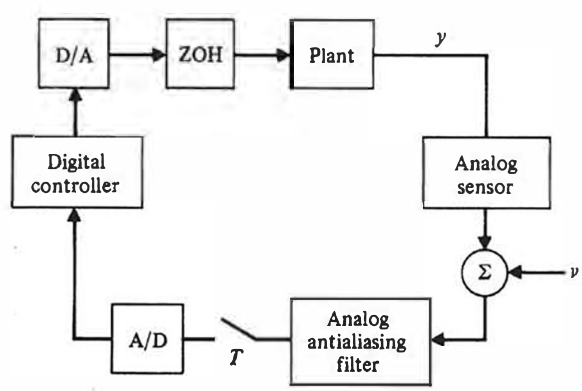


Figure 11.11
Block diagram showing the location of the antialiasing filter

digitized signal directly. Antialiasing filters are low pass, and the simplest transfer function is[5]

$$G_p(s) = \frac{\omega_p}{s + \omega_p}, \tag{11.16}$$

so that the noise above the prefilter breakpoint [ω_p in Eq. (11.16)] is attenuated. The design goal is to provide enough attenuation at half the sample rate ($\omega_s/2$) so that the noise above $\omega_s/2$, when aliased into lower frequencies by the sampler, will not be detrimental to the control-system performance. The basic idea of aliasing was discussed in Section 5.2 and shown to be capable of transforming a very-high-frequency noise into a frequency that is well within the bandwidth of the control system, thus allowing the system to respond to it.

◆ **Example 11.6** *Effect of Prefiltering on Aliasing*

For a 1-Hz sine wave with a 60-Hz sine wave superimposed to represent noise, find a prefilter that will eliminate distortions when sampled at 28 Hz. Demonstrate the effect of the prefilter, and lack of one, by plotting the signals over 2 cycles.

Solution. Figure 11.12(a) shows the 1-Hz sine wave with a 60-Hz sine wave superimposed to represent measurement noise. If this analog signal is sampled as is, the high-frequency noise will be aliased; Fig. 11.12(b) shows the results for sampling at $\omega_s = 28$ Hz and we see that the 60-Hz noise has been changed to a much lower frequency and appears as a distortion of the original sine wave. Fig. 11.12(c) shows the results of passing the noisy signal in (a) through a first order antialiasing filter with a breakpoint, $\omega_p = 20$ rad/sec (3.2 Hz). The breakpoint was picked considerably below the noise frequency so there is a large attenuation of the noise; and yet sufficiently above the 1-Hz signal so it was not attenuated. Sampling this clean signal results in the faithful reproduction of the signal shown in Fig. 11.12(d).

For an analog control system with a bandwidth on the order of 1 Hz, the 60-Hz noise would be too fast for the system to respond, and the noise would not be apparent on the system output. For a 1-Hz bandwidth digital control system without an analog prefilter, aliasing of the noise to a lower frequency as shown in Fig. 11.12(b) would allow the system to respond, thus producing sensitivity to errors that do not exist for analog controllers. Thus we see the reason for using antialiasing filters. ◆

To study the effect of sensor noise on sample-rate selection, we use the same analysis procedures that were used for the random plant disturbances.

5 Higher-order filters are also used in order to obtain better high-frequency attenuation with minimal low-frequency phase lag; common filter types are Bessel, Butterworth, and ITAE. See Franklin, Powell, and Emami-Naeini (2019) or Åström and Wittenmark (1997) for details.

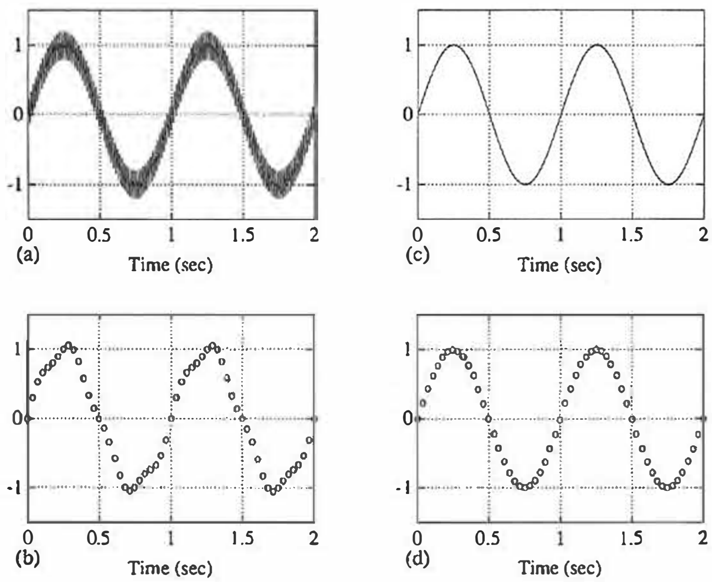


Figure 11.12
Demonstration of the effects of an antialiasing filter or prefilter, Example 11.6. (a) Signal plus noise; (b) samples of (a) at $\omega_s = 28$ Hz; (c) signal in (a) passed through antialiasing filter; (d) sampling of signal in (c)

Equation (11.5) is used to determine the system output errors due to continuous-measurement noise acting on a continuous controller for comparison purposes, and then Eq. (11.6) is used to determine the degradation for digital controllers with various sample rates. The only difference from the cases examined in Section 11.3 is that the noise to be examined enters on the plant output instead of on the plant input. A complication arises, however, because the prefilter breakpoint, ω_p, is a design variable whose selection is intimately connected with the sample rate. It is therefore necessary to examine both quantities simultaneously.

A conservative design procedure is to select the breakpoint and ω_s sufficiently higher than the system bandwidth so that the phase lag from the prefilter does not significantly alter the system stability; thus the prefilter can be ignored in the basic control system design. Furthermore, for a good reduction in the high-frequency noise at $\omega_s/2$, the sample rate should be selected about five times higher than the prefilter breakpoint. The implication of this prefilter design procedure is that sample rates need to be on the order of 30 to 50 times faster than the system bandwidth. This kind of conservative prefilter and sample-rate design procedure is likely to suggest that the sample rate needs to be higher than the other factors discussed in this chapter.

An alternative design procedure, due to Peled (1978), is to allow significant phase lag from the prefilter at the system bandwidth and thus to require that the control design be carried out with the analog prefilter characteristics included. Furthermore, the analysis procedure described above is carried out to determine more precisely what the effect of sampling is on the system performance. This procedure allows us to use sample rates as low as 10 to 30 times the system bandwidth, but at the expense of increased complexity in the design procedure. In addition, some cases can require increased complexity in the control implementation to maintain sufficient stability in the presence of prefilter phase lag; that is, the existence of the prefilter might, itself, lead to a more complex digital control algorithm.

◆ **Example 11.7** *Measurement Noise Effects vs. Sample Rate and Prefilter Breakpoint*

Use the double integrator plant as in Examples 11.1, 11.2, and 11.3 to demonstrate the degradation of the digital system response vs. the continuous control for various sample rates and prefilter breakpoints. Assume that only x_1 is available for measurement and use an estimator to reconstruct the state. Use the antialiasing filter as shown in Eq. (11.16).

Solution. The results of the digital controllers with the various prefilter breakpoints, ω_p, were all normalized to the results from a continuous controller with no prefilter. The exact evaluation of the integral in Eq. (11.7) was required because the sample rate was on the same order as ω_p for some sample rates studied. The continuous system was designed using the LQR and LQE methods of Chapter 9 so that the control roots were at $s = -1.5 \pm j1.5$ rad/sec and the estimator roots were at $s = -3.3 \pm j3.3$ rad/sec. All the digital controllers had roots at the discrete equivalent of those s-plane locations by using $z = e^{sT}$. The results of the normalized rms values of the output error due to white continuous-measurement noise entering the system as shown in Fig. 11.11 are shown in Fig. 11.13 for four different values of the prefilter breakpoint.

Note in the figure that, for sampling multiples below $\omega_s/\omega_b = 40$, the performance improves as the prefilter breakpoint decreases, even though it includes the case where the breakpoint is only twice the bandwidth. If a system is dominated by measurement noise as in this example and the designer chooses to limit the discrete degradation to 20% compared to the continuous case, the figure shows that a sampling multiple of

$$\omega_s/\omega_b \geq 25 \tag{11.17}$$

is adequate providing the prefilter breakpoint ratio $\omega_p/\omega_b \leq 5$, whereas a sampling multiple of $\omega_s/\omega_b = 40$ is required if the more conservative prefilter breakpoint ratio of $\omega_p/\omega_b = 10$ is used. An even slower sample rate would be adequate if a slower breakpoint was acceptable.

Also note in the figure at the very fast sample rates ($\omega_s/\omega_b > 200$) that the rms response decreased as the prefilter breakpoint increased, the opposite trend to that at the lower sample rates. This observation has little effect on the overall design because all values were within 10% of the continuous case for all the fast sample rates. However, there are detrimental effects of prefilters that can be more serious (see Hirata, 1989), and their influence on random plant disturbances should also be evaluated according to Section 11.3 before a design is finalized.

◆

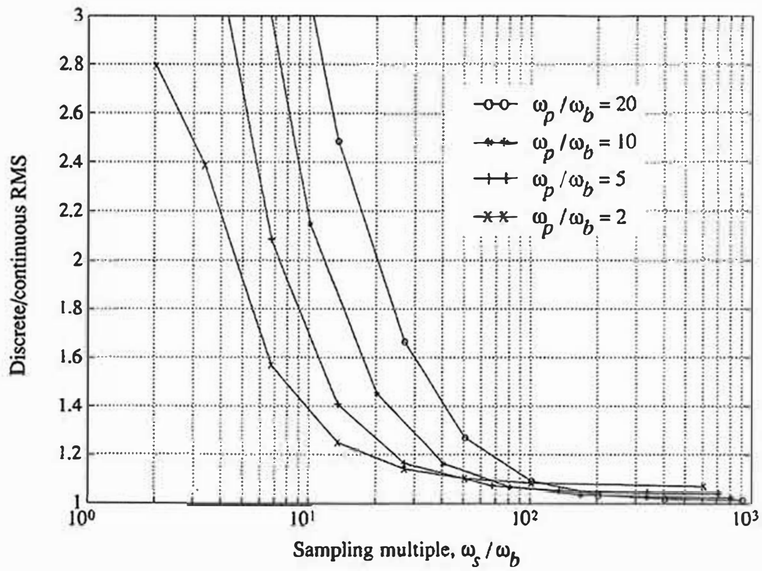


Figure 11.13
Root mean square response of Example 11.7 to white sensor noise showing effects of prefiltering, ω_p, and sampling, ω_s

In summary, there are two main conclusions. The first is that prefilters for removing high frequency measurement noise are effective and a methodology has been presented for their design. The second is that the bandwidth of the prefilter should be selected primarily for the reduction of sensor noise effects, and values approaching the system closed-loop bandwidth ($\omega_p/\omega_b \cong 2$) should be considered even though their phase lag needs to be adjusted for in the controller design. An exception to this would occur if the prefilter lag was sufficient to require a more complicated controller, because this increase in computation could offset any cost gains from slower sampling.

11.6 Multirate Sampling

In multi-input, multi-output (MIMO) systems, it is sometimes useful to use different sample rates on the various measurements and control variables. For example, in the tape-drive design in Chapter 9, one could logically sample the tension measurement, T_e in Fig. 9.18, faster than the tape position measurement, x_3, because the tension results from faster natural dynamics than the tape positioning. For MIMO systems with significant time constant differences in some natural modes or control loops, improvements in performance can be realized by sampling at different rates; such a system is referred to as a **multirate** (MR) system. In this section we will address the issues of analysis of an MR system and selection of the sample rates.

The z-transform method that is presented in Chapter 4 and is the basis for the analysis of discrete systems throughout this book does not apply to MR systems. Considerable research on the subject was carried out in the 1950s and resulted in the "switch decomposition" method, which has been reviewed by Ragazzini and Franklin (1958) and Amit (1980) and extended to multivariable systems by Whitbeck and Didaleusky (1980). The key idea in this method is that it is possible to analyze an MR system by reducing it to an equivalent single-rate system operating at the longest sample period of the system. The "Kranc" operator[6] facilitates this transformation.

Another approach uses a state-space description due to Kalman and Bertram (1959). The difference equations have coefficients in $\mathbf{\Phi}$ and $\mathbf{\Gamma}$ that change periodically at the longest sample period. An MR analysis procedure with this approach is contained in Berg, Amit, and Powell (1988) and, similarly to switch decomposition, reduces the MR system to an equivalent single-rate system where conventional methods can be employed. The computational burden for both the switch-decomposition and the state-space MR analysis is substantial; therefore, digital control designers typically opt for design methods during the synthesis process which bypass the need for using one of the exact MR analysis techniques.

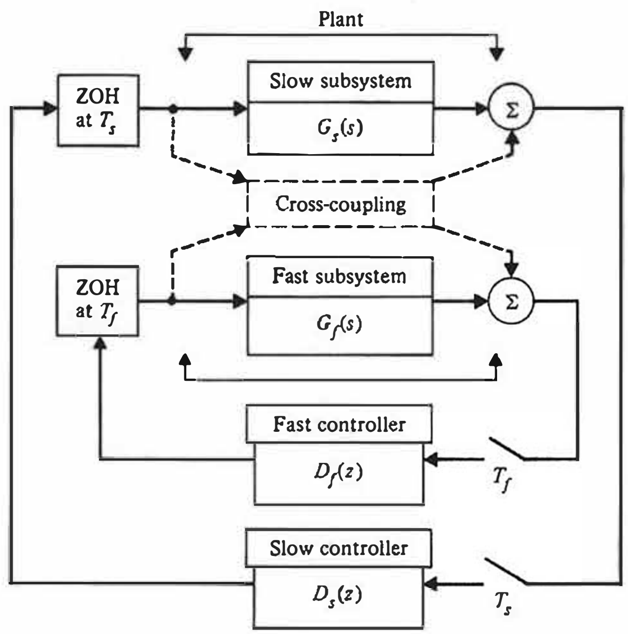
successive loop closure

An analysis method often used is called **successive loop closure** (SLC). It requires that the system initially be decoupled into two or more single-rate subsystems along the lines discussed in Section 9.1 and as shown in Fig. 11.14. The dashed lines in the figure show the cross-coupling between $G_f(s)$ and $G_s(s)$, which is ignored in the decoupling process. The controller for the subsystem with the highest frequencies (and fastest sample rate) is designed first with the sample rate selected according to the ideas in this chapter. This digitally controlled subsystem is then converted back to a continuous system (see d2c.m in MATLAB), which makes it possible to meld it into the slow portion of the plant. This transformation is the inverse of Eqs. (4.41) or (4.58) and is exact in the sense that the continuous model outputs will match the discrete system at the sample times. A discrete representation at the slow sample rate of the entire plant, including the previously ignored cross-coupling and the fast controller, can now be obtained using conventional techniques. This enables design of a digital controller for the slow loop using the resulting discrete model. This model is exact provided the fast sample rate is an integer multiple of the slow rate.

The approximation made during the design of the inner (fast) loop is the elimination of the cross-coupling; however, the model of the system at the slow sample rate will accurately predict any inner-loop root migration due to the approximation and can be corrected by a second design iteration, if needed.

6 See Kranc (1957).

Figure 11.14
Block diagram of a MIMO system showing methodology for multirate analysis using successive loop closure



Therefore, we see that the final design of the MR system should perform according to that predicted by linear analysis. The effect of the approximations made for design of the inner loop serve only to produce an inner-loop controller that may be somewhat inferior to that which could be obtained with an accurate MR analysis and optimal design of the entire system together as described by Berg, Amit, and Powell (1988).

For systems with three or more identifiable loops where it is desirable to have three or more different sample rates, the same ideas are invoked. The innermost (fastest) loop is closed first, transformed to a continuous model followed by transformation to a discrete model at the next slower sample rate, and so on until the final (slowest) loop is closed, encompassing the entire system.

The sample-rate selection for an MR system designed using SLC is carried out using the considerations discussed in Sections 11.1 through 11.5. Note that this does not imply that the same sampling multiple, ω_s/ω_b, be used for all loops. The appropriate sampling multiple is a function of the plant disturbance and sensor noise acting on that portion of the system which may be quite different for the different loops. Ignoring noise effects leads one to pick the same sampling multiple for all loops; that is, the sample rates for each loop are in the same proportion as the bandwidth for each loop.

◆ **Example 11.8** *Advantage of MR Sampling for the Double Mass-Spring System*

Investigate the merits of multirate sampling in terms of rms response to plant noise for the double mass-spring system.[7] Compare the rms response of an MR controller with that of a single rate (SR) controller that uses the same amount of computation effort.

Solution. The double mass–spring system with control applied to both masses is shown in Fig. 11.15. We have defined x_1 as the absolute position of the large mass, whereas x_2 is the position of the small mass with respect to the large mass. The inner loop consists of the measurement of x_2 and its feedback through a controller to u_2. The effect of the coupling from x_1 motion is ignored; so the dynamics are that of a simple mass-spring system. The inner (fast) loop was designed using pole-placement for full state feedback so that its open-loop resonant frequency of 4 Hz was changed to a closed-loop frequency of 8 Hz, and the damping improved from $\zeta = 0$ to $\zeta = 0.7$. The sample rate was chosen to be 71 Hz, nine times faster than the closed-loop root.

A continuous model of this fast inner loop was then obtained via d2c.m in MATLAB and added to the dynamics of the large mass, including the coupling between the two masses. The entire system was then discretized at the slow rate. The outer loop was designed using feedback from the slow sampled states so that the rigid-body open-loop roots at $s = 0$ were changed to 1 Hz with a damping of $\zeta = 0.7$. The sample rate for this loop was chosen to be 9 Hz, also nine times the closed-loop root. The effect of the dynamics of the large mass and its coupling to the small mass along with the outer-loop feedback caused an insignificant movement of the closed-loop inner roots. If the inner roots had moved an unacceptable amount, it would have been necessary to return to the inner-loop design and revise the design so that it resulted in acceptable dynamics *after* the second loop closure.

For evaluation of the MR controller, its performance in the presence of plant disturbance noise, w, applied to the small mass as shown in Fig. 11.15 will be compared against the performance of a single-rate controller. The sample rate of the single-rate controller was selected to be 40 Hz, a value that required approximately the same amount of computer computation time as the MR controller, thus yielding a fair comparison. Fig. 11.16 shows the relative rms responses for the two cases and generally shows the superiority of the MR case in that its small mass response was approximately 30% less than the single-rate case, whereas the large mass response was approximately the same. One could say that, for this configuration, it pays to assign a large fraction of the computer power on the control of the small mass by

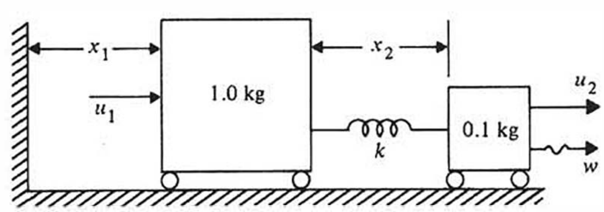


Figure 11.15 MIMO double mass–spring system used in Example 11.8

7 This example is from Berg, Amit, and Powell (1988).

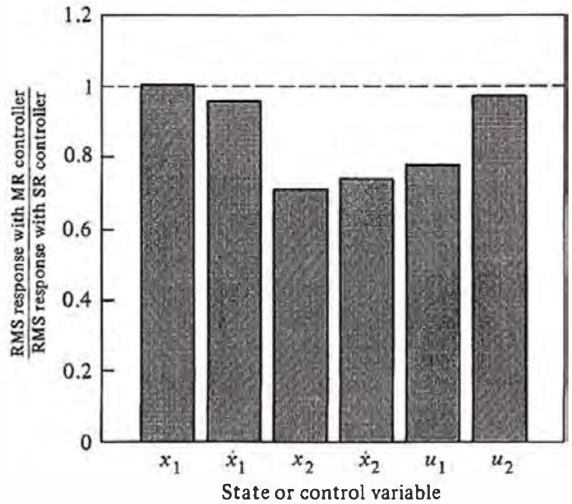


Figure 11.16
Relative rms state and control responses for Example 11.8 with *w* applied to the small mass

employing a faster sample rate for the small mass feedback. This follows because the bandwidth of the inner loop was faster and the plant disturbance noise was applied to this mass.

To analyze the situation further, let us change the plant disturbance noise configuration from the small mass to the large mass and keep the controller described above. The same comparison of relative rms responses is shown in Fig. 11.17 and indicates a different conclusion. The degradation of the large mass response with the MR controller is more than the improvement of the small mass; therefore, one could conclude that the single-rate controller was superior in this configuration. The reason is that the noise applied to the large mass (slow

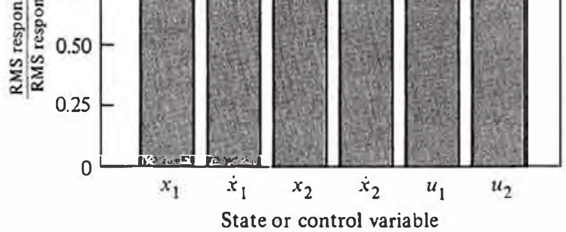


Figure 11.17
Relative rms state and control responses for Example 11.8 with *w* applied to the large mass

mode) produced a situation where the overall system response was improved by shifting more computer power to the slow mode compared to the previous configuration. The net result was that an equal sample rate on the two loops was best in spite of the fact that the roots of the two loops differed by a ratio of 8:1.

The significant benefit of the MR sampling shown in Fig. 11.16 is partially due to the fact that both sample rates were relatively slow, that is, nine times their respective closed-loop root. Had the sample rates been faster, the advantage would have been less pronounced. The reason for this is that the effect of disturbance noise versus sample rate is basically a quadratic function, with the errors growing quickly as sampling multiples fall below about 20. If all rates had been faster, the system performance would be less sensitive to both sample rates and thus less sensitive to the allocation of computer power to the fast or slow loop.

◆

In summary, we see that multirate sampling allows for the assignment of available computer power to the control modes that can most benefit, thus improving system response. The design of multirate systems can usually be carried out using a successive loop closure method that employs analytical techniques, which are considerably simpler than exact MR techniques. The resulting design is typically acceptable.

Multirate sampling is not necessary in the majority of digital control systems. In the usual case, sufficient computer speed is available so that one sample rate based primarily on the fastest mode can be selected even though one or more of the slower modes will be sampled faster than necessary. This is a better overall design because it avoids the complication of multirate sampling, even though it could be claimed that computer power is being wasted on some of the slower modes.

11.7 Summary

- For a control system with a specified closed-loop bandwidth, ω_b, the theoretical lower bound on the sample rate to bandwidth multiple is

$$\omega_s/\omega_b > 2.$$

Other considerations usually dictate that the sample rate be considerably faster than 2 times the closed-loop bandwidth.

- In order to provide a "reasonably smooth" control response, a sampling multiple of

$$\omega_s/\omega_b > 20$$

is often judged desirable. This is a subjective issue and depends heavily on the specific application, with slower rates often acceptable.

- There is potentially a full sample period delay between command input and the system response. In order to limit the possible delay to be 10% of the rise time, a sampling multiple of

$$\omega_s/\omega_b > 20$$

  is required.
- The primary purpose of many feedback control systems is to limit the response to random plant disturbances. In comparison to a continuous control system with the same equivalent s-plane roots, a digital control system degrades as the sample rate decreases. This degradation is shown by the bottom curve in Fig. 11.6 for a double integrator plant. In order to keep the degradation to be within 20% of a continuous controller, a sampling multiple of

$$\omega_s/\omega_b > 20$$

  is required.
- For a system with a resonance at ω_r that is being damped or stabilized by the controller, it is prudent to select a sample rate

$$\omega_s > 2\omega_r.$$

- For a system with a resonance at ω_r that is being damped or stabilized by the controller, it may be useful for robustness purposes to sample considerably faster than $2\omega_r$.
- **Analog prefilters** or **antialiasing filters** are typically required to attenuate the effect of measurement noise at frequencies greater than $\omega_s/2$.
- Selection of the prefilter breakpoint, ω_p, is a complex design step that involves interactions with the sample rate and control system bandwidth. Generally, the further the breakpoint is below $\omega_s/2$, the better the measurement noise attenuation. It is also convenient to maintain the prefilter breakpoint above the control system bandwidth; however, this is of lessor importance. The tradeoffs for a double integrator plant are depicted in Fig. 11.13. It shows that a sampling multiple of

$$\omega_s/\omega_b > 25$$

  is required for a breakpoint ratio of $\omega_p/\omega_b = 5$ if the designer wishes to limit the discrete noise degradation to 20% over that from the continuous case. Faster antialiasing filters require faster sample rates. Slower antialiasing filters reduce the effect of measurement noise, but more phase lag is introduced into the feedback and, if excessive, may necessitate extra lead in the controller.
- When using a small word size microprocessor (e.g., 8 bits), sampling multiples greater than 50 were shown in Fig. 11.6 to be counterproductive.

- The performance of a digital control system can sometimes be enhanced by using more than one sample rate for different loops in the system. The z-transform does not apply directly to this case and a successive loop closure design method for such systems was described in Section 11.6.
- The cost of a digital control system increases with sample rate due to the microprocessor cost and A/D cost. Therefore, for high volume products where the unit hardware costs dominate, picking the slowest sample rate that meets the requirements is the logical choice. For a low volume product, the labor cost to design the controller far outweighs the extra cost of a fast computer; therefore the logical choice of the sampling multiple is on the order of 40 or more so that the extra complexity of digital design methods is not required.

11.8 Problems

11.1 Find discrete compensation for a plant given by

$$G(s) = \frac{0.8}{s(s+0.8)}.$$

Use s-plane design techniques followed by the pole-zero mapping methods of Chapter 6 to obtain a discrete equivalent. Select compensation so that the resulting undamped natural frequency (ω_n) is 4 rad/sec and the damping ratio (ζ) is 0.5. Do the design for two sample periods: 100 ms and 600 ms. Now use an exact z-plane analysis of the resulting discrete systems and examine the resulting root locations. What do you conclude concerning the applicability of these design techniques versus sample rate?

11.2 The satellite attitude-control transfer function (Appendix A) is

$$G_1(z) = \frac{T^2(z+1)}{2(z-1)^2}.$$

Determine the lead network (pole, zero, and gain) that yields dominant poles at $\zeta = 0.5$ and $\omega_n = 2$ rad/sec for $\omega_s = 1, 2,$ and 4 Hz. Plot the control and output time histories for the first 5 sec and compare for the different sample rates.

11.3 Consider the satellite design problem of Appendix A.

(a) Design full state feedback controllers with both roots at $z = 0$ for $T = 0.01, 0.1,$ and 1 sec.

(b) Plot the closed-loop magnitude frequency response of θ for the three sample rates and determine the closed-loop bandwidth for each case.

(c) Plot the step response for the three sample rates and determine the rise time. Compare these values against that predicted by the relation in Section 2.1.7, $t_r \cong 1.8/\omega_n$.

11.4 Following the satellite design problem used for Example 11.3, evaluate the rms errors versus sample rate due to plant disturbance noise, using the case where only the attitude, θ, is measured and used by an estimator to reconstruct the state for control. Base all discrete control gains on z-plane roots that are equivalent to $\zeta = 0.7$ and $\omega_n = 3$ rad/sec and estimator gains on z-plane roots equivalent to $\zeta = 0.7$ and $\omega_n = 6$ rad/sec. Assume the torque disturbance has a power spectral density of $R_{w_{psd}} = 1$ N^2–m^2/sec as was done for Example 11.3. Plot the error for $\omega_s = 6$, 10, 20, 40, 80, and 200 rad/sec and comment on the selection of T for disturbance rejection in this case.

11.5 In Appendix A, Section A.4, we discuss a plant consisting of two coupled masses. Assume $M = 20$ kg, $m = 1$ kg, $k = 144$ N/m, $b = 0.5$ N-sec/m, and $T = 0.15$ sec.

(a) Design an optimal discrete control for this system with quadratic cost $\mathcal{J} = y^2 + u^2$. Plot the resulting step response.

(b) Design an optimal discrete filter based on measurements of d with $R_v = 0.1$ m^2 and a disturbing noise in the form of a force on the main mass M with $R_w = 0.001$ N^2.

(c) Combine the control and estimation into a controller and plot its frequency response from d to u.

(d) Simulate the response of the system designed above to a step input command and, leaving the controller coefficients unchanged, vary the value of the damping coefficient b in the plant. Qualitatively note the sensitivity of the design to this parameter.

(e) Repeat steps (a) to (d) for $T = 0.25$ sec.

11.6 For each design in Problem 11.2, increase the gain by 20% and determine the change in the system damping.

11.7 Consider a plant consisting of a diverging exponential, that is,

$$\frac{x(s)}{u(s)} = \frac{a}{s-a}.$$

Controlled discretely with a ZOH, this yields a difference equation, namely,

$$x(k+1) = e^{aT}x(k) + [e^{aT} - 1]u(k).$$

Assume proportional feedback,

$$u(k) = -Kx(k),$$

and compute the gain K that yields a z-plane root at $z = e^{-bT}$. Assume $a = 1$ sec^{-1}, $b = 2$ sec^{-1}, and do the problem for $T = 0.1$, 1.0, 2, 5 sec. Is there an upper limit on the sample period that will stabilize this system? Compute the percent error in K that will result in an unstable system for $T = 2$ and 5 sec. Do you judge that the case when $T = 5$ sec is practical? Defend your answer.

11.8 In the disk drive servo design (Problem 9.11 in Chapter 9), the sample rate is often selected to be twice the resonance ($\omega_s = 6$ kHz in this case) so that the notch at the Nyquist rate ($\omega_s/2$) coincides with the resonance in order to alleviate the destabilizing effect of the resonance. Under what conditions would this be a bad idea?

11.9 For the MIMO system shown in Fig. 11.15, let $k = 10$ N/m. Design an MR sampling controller where the fast sampling rate on the x_2 to u_2 loop is 10 Hz ($T = 100$ msec) and the slow sampling on the x_1 to u_1 loop is 2 Hz ($T = 500$ msec). Pick the fast poles at 1.5 Hz with $\zeta = 0.4$ and the slow poles at 0.3 Hz with $\zeta = 0.7$.

11.10 For your design in Problem 11.9, evaluate the steady state rms noise response of x_1 and x_2 for an rms value of $w = 0.1$ N. Repeat the design with the same pole locations, for two cases:

(a) fast sample period $T_{fast} = 50$ msec, and slow sample period $T_{slow} = 500$ msec,

(b) fast sample period $T_{fast} = 100$ msec, and slow sample period $T_{slow} = 250$ msec, and evaluate the rms response in both cases. If extra computational capability existed, where should it be used? In other words, which sample rate should be increased?

11.11 Replot the information in the lower curve in Fig. 11.6 (that is, no quantization noise) for three cases: (a) process noise and no measurement noise (the case already plotted), (b) process noise and measurement noise with an rms = 1, and (c) measurement noise with an rms = 1 and no process noise. Use exactly the same controller and estimator as the current plot. Note that FIG116.M assumes $R_v = 1$ for purposes of computing the continuous estimator gains; however, in computing the rms response, no measurement noise was included.

· 12 ·

System Identification

A Perspective on System Identification

In order to design controls for a dynamic system it is necessary to have a model that will adequately describe the system's motion. The information available to the designer for this purpose is typically of two kinds. First, there is the knowledge of physics, chemistry, biology, and the other sciences which have over the years developed equations of motion to explain the dynamic response of rigid and flexible bodies, electric circuits and motors, fluids, chemical reactions, and many other constituents of systems to be controlled. However, it is often the case that for extremely complex physical phenomena the laws of science are not adequate to give a satisfactory description of the dynamic plant that we wish to control. Examples include the force on a moving airplane caused by a control surface mounted on a wing and the heat of combustion of a fossil fuel of uncertain composition. In these circumstances, the designer turns to data taken from experiments directly conducted to excite the plant and measure its response. The process of constructing models from experimental data is called **system identification**. In identifying models for control, our motivation is very different from that of modeling as practiced in the sciences. In science, one seeks to develop models of nature as it is; in control one seeks to develop models of the plant dynamics that will be adequate for the design of a controller that will cause the actual dynamics to be stable and to give good performance.

The initial design of a control system typically considers a small signal analysis and is based on models that are linear and time-invariant. Having accepted that the model is to be linear, we still must choose between several alternative descriptions of linear systems. If we examine the design methods described in the earlier chapters, we find that the required plant models may be grouped in two categories: parametric and nonparametric. For design via root locus or pole assignment, we require a parametric description such as a transfer function or a state-variable description from which we can obtain the poles and zeros of the plant. These equivalent models are completely described by the numbers that specify the coefficients of the polynomials, the elements of the state-description matrices, or the numbers that specify the poles and zeros. In either case we call these numbers the *parameters* of the model, and the category of such models is a **parametric description** of the plant model.

parametric model

In contrast to parametric models, the frequency-response methods of Nyquist, Bode, and Nichols require the curves of amplitude and phase of the transfer function $G(e^{j\omega T}) = Y(j\omega)/U(j\omega)$ as functions of ω. Clearly, if we happen to have a parametric description of the system, we can compute the transfer function and the corresponding frequency response. However if we are given the frequency response or its inverse transform, the impulse response, without parameters (perhaps obtained from experimental data) we have all we need to design a lead, lag, notch, or other compensation to achieve a desired bandwidth, phase margin, or other frequency response performance objective without ever knowing what the parameters are. We call the functional curves of $G(e^{j\omega T})$ a **nonparametric** model because in principle there is no finite set of numbers that describes it exactly.

nonparametric model

In addition to the selection of the type of model desired, one must define the error to be minimized by the model estimate. For transfer functions, if we assume the true value is $G(e^{j\omega T})$ and the model is given by $\hat{G}(e^{j\omega T})$, then one possible measure is $\int (G(e^{j\omega T}) - \hat{G}(e^{j\omega T}))^2 d\omega$. Clearly other measures are also possible and can be considered. For discrete-time parametric models a standard form assumes that the output is generated by the discrete equation

$$a(z)y = b(z)u + c(z)v, \tag{12.1}$$

where a, b, and c are polynomials in the shift operator[1] z, y is the output, u is the input, and v is the error and assumed to be a random process independent of u. If $c(z) = a(z)$, then v is called the output error, for in that case Eq. (12.1) becomes $y = [b(z)/a(z)]\ u + v$, and v is noise added to the output of the system. If $c(z) = 1$ then v is called the equation error, for it is the error by which the data y and u fail to satisfy the equation given by $a(z)y = b(z)u$. Finally, if $c(z)$ is a general polynomial, then Eq. (12.1) is equivalent to a Kalman filter and v is called the prediction error.

The data used in system identification consist mainly of the records of input, $u(k)$, and output, $y(k)$ and assumptions about the error. However, in addition to these records there is usually very important information about the physical plant that must be used to get the best results. For example, the plant transfer function may have a known number of poles at $z = 1$ corresponding to rigid body motion. If there are such known components, the data can be filtered so that the identification needs only estimate the unknown part. Also typically it

1 Some authors use the variable q as the shift operator to avoid confusion with z as the variable of the $\mathcal{Z}$ transform. It is expected that the rare use of z as an operator will not cause confusion here.

is known that the important features of the transfer function are contained in a known frequency range and that data outside that range is only noise. It is important to filter the raw data to remove these components in so far as possible without distorting important features to prevent the algorithm from distorting the estimated model in a futile effort to explain this noise. The important conclusion is that all *a priori* information be used fully to get the best results from the computations.

In summary, identification is the process by which a model is constructed from prior knowledge plus experimental data and includes four basic components: the data, the model set, the criterion by which one proposed model is to be compared with another and the process of validation by which the model is confirmed to mimic the plant on data not used to construct the model. Because of the large data records necessary to obtain effective models and the complexity of many of the algorithms used, the use of computer aids is essential in identification. Developments such as the MATLAB System Identification Toolbox are enormous aids to the practical use of the techniques described in this chapter.

Chapter Overview

In this chapter, we consider several of the most common and effective approaches to identification of linear models. In Section 1 we consider the development of linear models as used in identification. In Section 2 the construction of nonparametric models is considered. Data taken by single sinusoidal signal inputs as well as data in response to a complex signal are considered. Also presented is the concept of including prior knowledge of the plant response, including the possibility of known pole locations for rigid body cases. In Section 3 techniques for identification of parametric linear models based on equation error are introduced with the selection of parameters and definition of the error. Algorithms for batch least squares are given in Section 4, recursive least squares in Section 5, and stochastic least squares in Section 6. It is shown that bias is encountered when the equation error is not a white noise process in each of the techniques. In Section 7 the method of maximum likelihood is introduced as a method that can overcome the bias considering the prediction error. Finally, in Section 8 the estimation of state description matrices based on subspace methods is described.

12.1 Defining the Model Set for Linear Systems

Formally, one proceeds as follows with the process of linearization and small-signal approximations. We begin with the assumption that our plant dynamics are adequately described by a set of ordinary differential equations in state-variable form as discussed in Section 4.3.6

$$
\begin{aligned}
\dot{x}_1 &= f_1(x_1, \ldots, x_n, u_1, \ldots, u_m, t), \\
\dot{x}_2 &= f_2(x_1, \ldots, x_n, u_1, \ldots, u_m, t), \\
&\vdots \\
\dot{x}_n &= f_n(x_1, \ldots, x_n, u_1, \ldots, u_m, t), \\
y_1 &= h_1(x_1, \ldots, x_n, u_1, \ldots, u_m, t), \\
&\vdots \\
y_p &= h_p(x_1, \ldots, x_n, u_1, \ldots, u_m, t),
\end{aligned}
\tag{12.2}
$$

or, more compactly in matrix notation, we assume that our plant dynamics are described by

$$
\begin{aligned}
\dot{\mathbf{x}} &= \mathbf{f}(\mathbf{x}, \mathbf{u}, t), \qquad \mathbf{x}(t_0) = \mathbf{x}_0, \\
\mathbf{y} &= \mathbf{h}(\mathbf{x}, \mathbf{u}, t).
\end{aligned}
\tag{12.3}
$$

The assumption of stationarity is here reflected by the approximation that $\mathbf{f}$ and $\mathbf{h}$ do not change significantly from their initial values at t_0. Thus we can set

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t_0)$$

or, simply,

$$
\begin{aligned}
\dot{\mathbf{x}} &= \mathbf{f}(\mathbf{x}, \mathbf{u}), \\
\mathbf{y} &= \mathbf{h}(\mathbf{x}, \mathbf{u}).
\end{aligned}
\tag{12.4}
$$

The assumption of small signals can be reflected by taking $\mathbf{x}$ and $\mathbf{u}$ to be always close to their reference values $\mathbf{x}_0$, $\mathbf{u}_0$, and these values, furthermore, to be a solution of Eq. (12.4), as

$$\dot{\mathbf{x}}_0 = \mathbf{f}(\mathbf{x}_0, \mathbf{u}_0) \tag{12.5}$$

Now, if $\mathbf{x}$ and $\mathbf{u}$ are close to $\mathbf{x}_0$ and $\mathbf{u}_0$, they can be written as $\mathbf{x} = \mathbf{x}_0 + \delta\mathbf{x}$; $\mathbf{u} = \mathbf{u}_0 + \delta\mathbf{u}$, and these can be substituted into Eq. (12.4). The fact that $\delta\mathbf{x}$ and $\delta\mathbf{u}$ are small is now used to motivate an expansion of Eq. (12.4) about $\mathbf{x}_0$ and $\mathbf{u}_0$ and to suggest that only the terms in the first power of the small quantities $\delta\mathbf{x}$ and $\delta\mathbf{u}$ need to be retained. We thus have a vector equation and need the expansion of a vector-valued function of a vector variable

$$\frac{d}{dt}(\mathbf{x}_0 + \delta\mathbf{x}) = \mathbf{f}(\mathbf{x}_0 + \delta\mathbf{x}, \mathbf{u}_0 + \delta\mathbf{u}). \tag{12.6}$$

If we go back to Eq. (12.2) and do the expansion of the components f_i one at a time, it is tedious but simple to verify that Eq. (12.6) can be written as

$$\dot{\mathbf{x}}_0 + \delta\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}_0, \mathbf{u}_0) + \mathbf{f}_{\mathbf{x}}(\mathbf{x}_0, \mathbf{u}_0)\delta\mathbf{x} + \mathbf{f}_{\mathbf{u}}(\mathbf{x}_0, \mathbf{u}_0)\delta\mathbf{u} + \cdots \tag{12.7}$$

where we define the partial derivative of a scalar f_i with respect to the vector $\mathbf{x}$ by a subscript notation

$$f_{1,\mathbf{x}} \triangleq \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \end{pmatrix}. \tag{12.8}$$

The row vector in Eq. (12.8) is called the *gradient* of the scalar f_1 with respect to the vector $\mathbf{x}$. If $\mathbf{f}$ is a vector, we define its partial derivatives with respect to the vector $\mathbf{x}$ as the matrix (called the Jacobean) composed of *rows* of gradients. In the subscript notation, if we mean to take the partial of *all* components, we use the bold vector as the subscript

$$\mathbf{f}_{\mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial \mathbf{x}} \\ \frac{\partial f_2}{\partial \mathbf{x}} \\ \vdots \\ \frac{\partial f_n}{\partial \mathbf{x}} \end{bmatrix}. \tag{12.9}$$

Now, to return to Eq. (12.7), we note that by Eq. (12.5) we chose $\mathbf{x}_0$, $\mathbf{u}_0$ to be a solution so the first terms of Eq. (12.7) balance, and, because the terms beyond those shown depend on higher powers of the small signals $\delta\mathbf{x}$ and $\delta\mathbf{u}$, we are led to the approximation

$$\begin{aligned} \delta\dot{\mathbf{x}} &\approx, \mathbf{f}_{\mathbf{x}}(\mathbf{x}_0, \mathbf{u}_0)\delta\mathbf{x} + \mathbf{f}_{\mathbf{u}}(\mathbf{x}_0, \mathbf{u}_0)\delta\mathbf{u} \\ \delta\mathbf{y} &= \mathbf{h}_{\mathbf{x}}\delta\mathbf{x} + \mathbf{h}_{\mathbf{u}}\delta\mathbf{u}. \end{aligned} \tag{12.10}$$

Now the notation is overly clumsy and we drop the δ-parts and define the constant matrices

$$\begin{aligned} \mathbf{F} &= \mathbf{f}_{\mathbf{x}}(\mathbf{x}_0, \mathbf{u}_0), & \mathbf{G} &= \mathbf{f}_{\mathbf{u}}(\mathbf{x}_0, \mathbf{u}_0) \\ \mathbf{H} &= \mathbf{h}_{\mathbf{x}}(\mathbf{x}_0, \mathbf{u}_0) & \mathbf{J} &= \mathbf{h}_{\mathbf{u}}(\mathbf{x}_0, \mathbf{u}_0) \end{aligned}$$

to obtain the form we have used in earlier chapters

$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} + \mathbf{G}\mathbf{u}, \qquad \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{J}\mathbf{u}. \tag{12.11}$$

We will go even further in this chapter and restrict ourselves to the case of single input and single output and discrete time. We write the model as

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{\Phi}\mathbf{x}(k) + \mathbf{\Gamma}u(k), \\ y(k) &= \mathbf{H}\mathbf{x}(k) + Ju(k), \end{aligned} \tag{12.12}$$

from which the transfer function is

$$G(z) = Y(z)/U(z) = \mathbf{H}(z\mathbf{I} - \mathbf{\Phi})^{-1}\mathbf{\Gamma} + J, \tag{12.13}$$

and the unit pulse response is

$$g(0) = J \qquad g(k) = \mathbf{H}\mathbf{\Phi}^{k-1}\mathbf{\Gamma} \qquad k = 1, \ldots \tag{12.14}$$

The System Identification Toolbox includes the multivariable case, and multivariable theory is covered in a number of advanced texts devoted to identification, including Ljung (1987).

12.2 Identification of Nonparametric Models

One of the important reasons that design by frequency-response methods is so widely used is that it is so often feasible to obtain the frequency response from experimental data. Furthermore, reliable experimental estimates of the frequency response can be obtained when noise is present and when the system is weakly nonlinear. Models based on such frequency-response data can then be used effectively in control system design using, for example, the methods of Bode. The presence of nonlinearity leads to the concept of the "describing function," a topic that will be considered in Chapter 13. Here we will introduce and illustrate experimental measurement of the frequency response in the case of linear, constant, stable models, including the possibility of unmeasured noise inputs to the system.

The situation to be considered is described by the transform equations

$$Y(z) = G(z)U(z) + H(z)W(z). \tag{12.15}$$

In this equation, Y is the plant output, U is the known plant control input, and W is the unmeasured noise (which might or might not be random). G is thus the plant transfer function and H is the unknown but stable transfer function from the noise to the system output. The frequency response of this system, as was discussed in Chapter 2, is the evaluation of $G(z)$ for z on the unit circle. If the noise is zero and the input $u(k)$ is a sinusoid of amplitude A and frequency ω_0, then $u(k)$ is given by

$$u(kT) = A \sin(\omega_0 kT), \tag{12.16}$$

and the output samples, in the steady state, can be described by

$$y(kT) = B \sin(\omega_0 kT + \phi_0), \tag{12.17}$$

where

$$\begin{aligned} B/A &= \left|G(e^{j\omega_0 T})\right| \\ \phi_0 &= \angle G(e^{j\omega_0 T}). \end{aligned} \tag{12.18}$$

identification, one-frequency-at-a-time

Given the situation described by Eq. (12.15), there are two basic schemes for obtaining a nonparametric estimate of $G(e^{j\omega T})$. The first of these can be described as the one-frequency-at-a-time method and the other as the spectral estimate method. We will describe these in turn. In each case we will require the computation of Fourier transforms using finite records of data; and for this the discrete Fourier transform (DFT) described in Chapter 4 is used, especially in its computationally optimized form, the fast Fourier transform (FFT).

The Method of One Frequency at a Time

We assume that we have a system as described in Eq. (12.15) and that the input is a sinusoid of the form of Eq. (12.16). The input and the output are recorded for N samples, and we wish to find the amplitude B and the phase ϕ_0, from which one point on the frequency response can be computed according to Eq. (12.18). We then increment the frequency to $\omega_1 = \omega_o + \delta\omega$ and repeat until the entire range of frequencies of interest is covered with the desired resolution. The accuracy of the estimate at any one point is a function of the number of points taken, the relative intensity of the noise, and the extent to which system transients have been allowed to decay before the recordings are started. We will describe the calculations in an ideal environment and discuss briefly the consequences of other situations.

Consider an output of the form of Eq. (12.17) but with unknown amplitude and phase. Recognizing that the experimental data might reflect some nonlinear effects and contain some noise component, a reasonable computational approach is to find that amplitude B and phase ϕ_0 which best fit the given data. We define the estimate of the output as

$$\begin{aligned}\hat{y}(kT) &= B\sin(\omega_0 kT + \phi_0)\\ &= B_c\cos(\omega_0 kT) + B_s\sin(\omega_0 kT),\end{aligned}$$

where $B_c = B\sin(\phi_o)$ and $B_s = B\cos(\phi_0)$. Once we have computed B_c and B_s we can compute the values of B and ϕ_0 as

$$\begin{aligned}B &= \sqrt{(B_c^2 + B_s^2)},\\ \phi_0 &= \arctan\frac{B_c}{B_s}.\end{aligned} \tag{12.19}$$

To find the best fit of $\hat{y}$ to the data, we form the quadratic cost in the error between y and $\hat{y}$ as

$$\begin{aligned}\mathcal{J} &= \frac{1}{N}\sum_{k=0}^{N-1}(y(kT) - \hat{y}(kT))^2\\ &= \frac{1}{N}\sum_{k=0}^{N-1}(y(kT) - B_c\cos(\omega_0 kT) - B_s\sin(\omega_0 kT))^2.\end{aligned}$$

Keep in mind that $y(kT)$ is the measured output and $\hat{y}(kT)$ is the computed estimate of the output. We wish to compute B_c and B_s so as to make $\mathcal{J}$ as small as possible. To do this, we set the derivative of $\mathcal{J}$ with respect to B_c and B_s equal to zero. Consider B_c first

$$\frac{\partial\mathcal{J}}{\partial B_c} = \frac{1}{N}\sum_{k=0}^{N-1}2(y(kT) - B_c\cos(\omega_0 kT) - B_s\sin(\omega_0 kT))(-\cos(\omega_0 kT)) = 0.$$

If we use elementary trigonometric identities for the product of sine and cosine, and collect terms of the sum, this expression is equivalent to

$$\frac{1}{N}\sum_{k=0}^{N-1} y(kT)\cos(\omega_0 kT) - \frac{B_c}{2} - \frac{B_c}{2N}\sum_{k=0}^{N-1}\cos(2\omega_0 kT)$$
$$-\frac{B_s}{2N}\sum_{k=0}^{N-1}\sin(2\omega_0 kT) = 0. \tag{12.20}$$

Generally, we would expect the last two terms in Eq. (12.20) to be small because the sinusoids alternate in sign and have zero average value. They will be *exactly* zero if we select the test frequencies carefully. Consider the following identity (which also played an important role in the development of the DFT in Chapter 4)

$$\frac{1}{N}\sum_{k=0}^{N-1} z^{-k} = \frac{1-z^{-N}}{N(1-z^{-1})}.$$

If we select integers ℓ and N and let $z = e^{j2\pi\ell/N}$, corresponding to $\omega = 2\pi\ell/N$, then

$$\frac{1}{N}\sum_{k=0}^{N-1} e^{-j2\pi\ell k/N} = \frac{1-e^{-j2\pi\ell}}{1-e^{-j2\pi\ell/N}}$$
$$= \begin{cases} 1, & \ell = rN \\ 0 & \text{elsewhere} \end{cases} \tag{12.21}$$

If we substitute Euler's formula, $e^{-j2\pi\ell k/N} = \cos(2\pi\ell k/N) - j\sin(2\pi\ell k/N)$, into the left side of Eq. (12.21) and equate real and imaginary parts, we find that

$$\frac{1}{N}\sum_{k=0}^{N-1}\cos(2\pi\ell k/N) = \begin{cases} 1 & \ell = rN \\ 0 & \text{elsewhere} \end{cases}$$

and also

$$\frac{1}{N}\sum_{k=0}^{N-1}\sin(2\pi\ell k/N) = 0.$$

If we now select our test frequencies to be $\omega_o = 2\pi\ell/NT$, for integer ℓ, then Eq. (12.20) reduces to

$$B_c = \frac{2}{N}\sum_{k=0}^{N-1} y(kT)\cos(2\pi\ell k/N). \tag{12.22}$$

In similar fashion, for this choice of input frequency, one can show that

$$B_s = \frac{2}{N}\sum_{k=0}^{N-1} y(kT)\sin(2\pi\ell k/N). \tag{12.23}$$

Thus Eq. (12.22) and Eq. (12.23) give us formulas for computing the values of B_c and B_s, and Eq. (12.19) gives us the formulas for the gain and phase that describe the output data $y(kT)$. The transfer function from which the output was measured is then given by Eq. (12.18).

Equation (12.22) is closely related to the DFT of y derived in Section 4.5. In fact, if we assume that $y(kT)$ is a sinusoid of frequency $\omega_\ell = 2\pi\ell/NT$ and consider the DFT of y

$$\begin{aligned} DFT[y(kT)] &= \sum_{k=0}^{N-1} y(kT)e^{-j2\pi kn/N} \\ &= \sum_{k=0}^{N-1} y(kT)[\cos(2\pi kn/N) - j\sin(2\pi kn/N)] \\ &= \begin{cases} \frac{N}{2}[B_c - jB_s], & n = \ell \\ 0 & n \neq \ell \end{cases} . \end{aligned}$$

Likewise, the DFT of the sinusoidal input of the same frequency is given by

$$\begin{aligned} DFT[u(kT)] &= \sum_{k=0}^{N-1} A\sin(2\pi\ell k/N)e^{-j2\pi\ell k/N} \\ &= \frac{A}{2j}\sum_{k=0}^{N-1}[e^{j2\pi\ell k/N} - e^{-j2\pi\ell k/N}]e^{-j2\pi nk/N} \\ &= \frac{NA}{2j}, \quad (n = \ell) \\ &= 0, \quad \text{elsewhere.} \end{aligned} \tag{12.24}$$

Thus, for $n = \ell$, the ratio of the DFT of y to that of u is

$$\begin{aligned} \frac{DFT(y)}{DFT(u)} &= \frac{(N/2)(B_c - jB_s)}{NA/2j} \\ &= \frac{B_s + jB_c}{A} \\ &= |G|e^{j\phi_0}, \end{aligned} \tag{12.25}$$

which is the result found in Section 4.5. Of course, using the DFT or the FFT on $y(kT)$ and $u(kT)$ just for one point on the transfer function is unnecessarily complicated, but because the DFT automatically computes the best fit of a sinusoid to the output in the least-squares sense, the result is very accurate and, with the use of the FFT, it is fast.

It is possible to take advantage of Eq. (12.25) to evaluate the transfer function at several frequencies at once by using an input that is nonzero over a band of frequencies and computing the ratio of output to input at each nonzero value. To illustrate this technique, we will use an input known as a *chirp*, which is a modulated sinusoidal function with a frequency that grows from an initial value to a final value so as to cover the desired range. An example of a chirp is given by the formulas

$$\begin{aligned} u(kT) &= A_0 + w(k)\sin(\omega_k kT), \qquad 0 \le k \le N-1, \\ w(k) &= A \,\mathrm{sat}(k/0.1N)\mathrm{sat}((N-k)/0.1N), \\ \omega_k &= \omega_{\text{start}} + \frac{k}{N}(\omega_{\text{end}} - \omega_{\text{start}}). \end{aligned} \tag{12.26}$$

In this expression, $w(k)$ is a weighting or window function that causes the input to start at zero, ramp up to amplitude A, and ramp down to zero at the end. The "sat" is the saturation function, which is linear from -1 to $+1$ and saturates at -1 for arguments less than -1 and at $+1$ for arguments greater than $+1$. The constant A_0 is added to make the input have zero average value.

◆ **Example 12.1** *A Chirp Signal*

Plot a chirp of maximum amplitude 1.0, starting frequency $\omega T = 0.1$ rad and ending frequency $\omega T = 0.75$ rad. Use 256 points. Also plot the magnitude of the transform of the signal.

fft

Solution. From Eq. (12.26) we have $A = 1$, $T\omega_{start} = 0.1$, $T\omega_{end} = 0.75$, and $N = 256$. The constant A_0 was computed as the negative of the average of the basic signal, using MATLAB. The transform was computed using fft.m. The results are plotted in Fig. 12.1, parts (a) and (b). Notice that the actual spectrum is very flat between 0.3 and 1.0 rad and has non-negligible energy over the wider range between 0.1 and 1.5 rad. The energy above $T\omega_{end} = 0.75$ rad is the result of the fact that the chirp is really a frequency modulated signal and many sidebands are present. ◆

◆ **Example 12.2** *Computing a Transfer Function*

Apply the chirp signal of Example 12.1 to a system obtained as the zero order hold equivalent of $1/(s^2 + 0.25s + 1)$ with $T = 0.5$. Compute the estimate of the transfer function using Eq. (12.25).

Solution. The results of the application of Eq. (12.25) to this data are plotted in Fig. 12.1, parts (c) and (d). The true frequency response is plotted as a solid line and the experimental transfer function estimate is plotted as x's on the same graphs. As can be seen, the estimate is a very accurate measure for frequencies up to 1.8 rad, after which the estimate deviates badly from the true magnitude and phase curves. This deviation coincides with the frequencies where the input has little energy and leads us to conclude that one should not trust the result of Eq. (12.25) unless the input has significant energy at the computed frequency. ◆

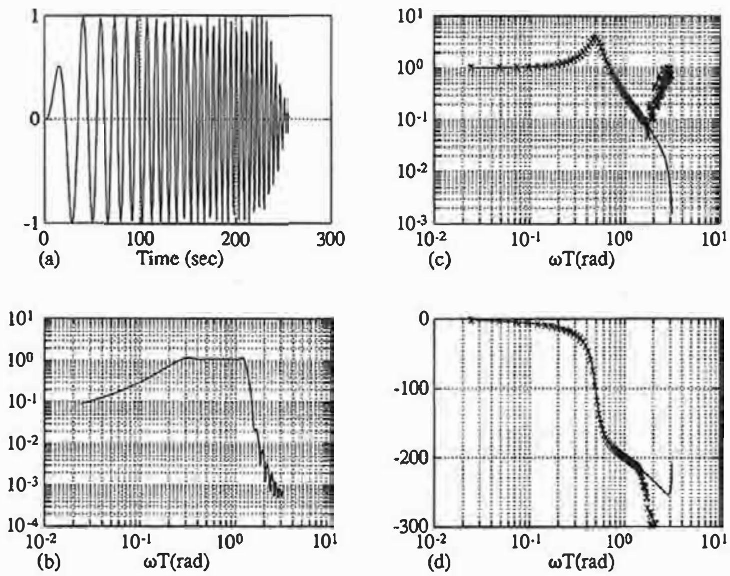


Figure 12.1
Estimation of transfer function using a chirp signal: (a) the chirp, (b) the chirp's spectrum, (c) the transfer function and its magnitude estimate, (d) the true and estimated phase

Data Conditioning

As illustrated by Example 12.1, a signal can be constructed to have energy only in a certain band, and with its use the transfer function can be accurately estimated in this band. This observation is a particular case of an extremely important principle in system identification:

> Use all available prior knowledge
> to reduce the uncertainty in the estimate.

In the case of Example 12.1, we were able to use the fact that the important features of the unknown transfer function are in the digital frequency range $0.1 \leq \omega T \leq 1.5$ to design a chirp signal that would provide energy in that range only. Another situation, which was developed by Clary (1984), occurs when part of the transfer function is already known. This is common in the control of mechanical devices, such as robot mechanisms or space structures, where one knows that the transfer function has two poles at the origin in the s-plane and thus two poles at $z = 1$ in the z-plane. We may also know how fast the transfer function goes to zero for high frequencies and from this knowledge assign zeros of the transfer function at $z = -1$ to correspond to these. This assumption is not exact, because zeros at infinity in the s-plane do not map to any fixed point in the z-plane; however, in almost all cases, they result in z-plane zeros that are often well outside the unit circle so that they have almost no influence on the frequency response over the important band and are very difficult to identify accurately. It is for this reason that known s-plane zeros at infinity can be assigned to $z = -1$ and used to reduce the uncertainty in the discrete transfer-function estimation.

Clary's method

To illustrate Clary's procedure, suppose we model the known poles as the roots of the polynomial $a_k(z)$ and model the known zeros as the roots of $b_k(z)$. Then the overall transfer function can be factored into known and unknown parts in the form

$$G(z) = \frac{b_k(z)}{a_k(z)} G_1(z). \tag{12.27}$$

In block-diagram form, the situation can be arranged as shown in Fig. 12.2. In this case, once the simpler unknown $G_1(z)$ is estimated from the filtered data u_f and y_f, the overall transfer function $G(z)$ can be reconstructed from Eq. (12.27).

◆ Example 12.3 *Using Known Components*

Consider the system formed by multiplying the plant of Example 12.2 by $1/s^2$. Identify the corresponding discrete system by Clary's method.

Solution. The resulting fourth-order discrete system has two poles at $z = 1$ and zeros at $z = -9.5$, $z = -0.98$, and $z = -0.1$. If the same chirp signal used in Example 12.2 is used to identify this transfer function from its input and output, the results are shown in Fig. 12.3(a) and (b). Clearly the estimates, shown by x's on the plots, are very poor in both magnitude and phase. However, if u is filtered by $(1 + z^{-1})^2$ and y is filtered by $(1 - z^{-1})^2$, then the transfer function G_1 can be estimated from these signals. Afterward, the estimated frequency

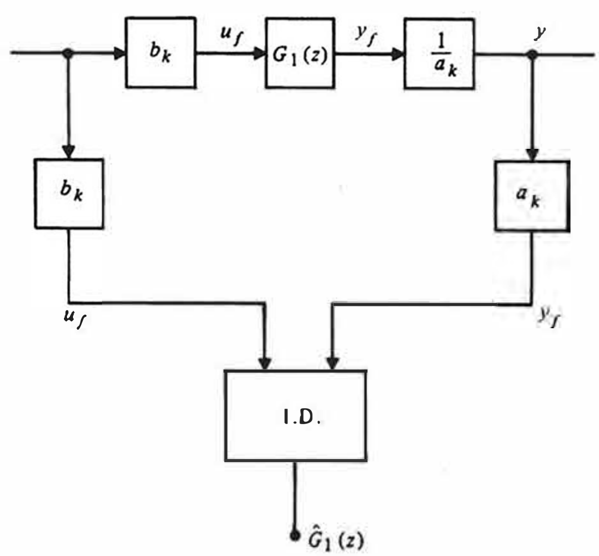


Figure 12.2
Block diagram of identification of a partially known transfer function

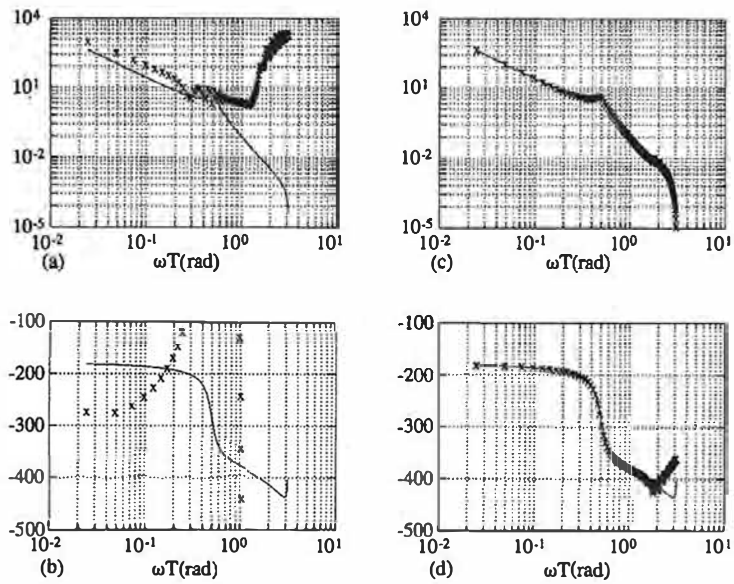


Figure 12.3
Transfer function estimates for a fourth-order system without filtering (a) and (b) and with filtering of the data, (c) and (d)

responses are multiplied by the known frequency responses according to Eq. (12.27) and the total frequency response is plotted in Fig. 12.3(c) and (d). Now an excellent estimate of the transfer function results.

Our final example of the use of prior information in identification occurs when it is known what the important frequency range is and, furthermore, that significant sensor noise appears in the plant output outside this range. Such noise is almost always present in experimental data and typically includes both noise from physical sources and noise that can be attributed to the quantization of A/D conversion. In this case, we observe that the identification of $G(z)$ is the same as the identification of $F(z)G(z)F(z)^{-1}$ because the filter transfer function F cancels. However, suppose that we filter both u and y through F. Although this has no influence on the relation between u and y, it can suppress the noise significantly. The equations that illustrate this effect are

$$\begin{aligned} y &= Gu + v, \\ Fy &= GFu + Fv, \\ y_f &= Gu_f + Fv. \end{aligned} \tag{12.28}$$

Therefore, we have succeeded in replacing the noise input v with the filtered noise Fv while keeping the same relation between the signal input and the output as before. If the noise includes significant energy at frequencies that are not important to the signal response, then F can be shaped to remove these components and greatly improve the transfer-function estimates. Most commonly, F is a lowpass filter such as a Butterworth characteristic with a cutoff frequency just above the range of important frequencies for the plant model. In the case of nonparametric estimation, the effect of the lowpass filter can be achieved simply by ignoring the estimates outside the desired range, as could be seen from Fig. 12.1. However, when we come to parametric estimation, the use of such filters will be found to be essential.

Stochastic Spectral Estimation

The deterministic methods just described require a high signal-to-noise ratio in the important band of the transfer function to be estimated. If this is not available, as frequently occurs, an alternative based on averaging or statistical estimation is available. The necessary equations are easily derived from the results on stochastic processes given in Appendix D. Suppose we describe the plant dynamics with the convolution

$$y(n) = \sum_{k=-\infty}^{\infty} g(k)u(n-k) \tag{12.29}$$

and assume that u and therefore y are zero mean stochastic processes. If we multiply Eq. (12.29) on both sides by $u(n-\ell)$ and take expected values, we obtain

$$\begin{aligned} \mathcal{E}y(n)u(n-\ell) &= \mathcal{E}\sum_{k=-\infty}^{\infty} g(k)u(n-k)u(n-\ell), \\ R_{yu}(\ell) &= \sum_{k=-\infty}^{\infty} g(k)R_{uu}(\ell-k), \end{aligned} \tag{12.30}$$

where $R_{yu}(\ell)$ is the cross-correlation between the y process and the u process and $R_{uu}(\ell-k)$ is the autocorrelation of the u process. If we take the z-transform of Eq. (12.30), it is clear, because this is a convolution, that

$$S_{yu}(z) = G(z)S_{uu}(z)$$

and thus that

$$G(z) = \frac{S_{yu}(z)}{S_{uu}(z)}, \tag{12.31}$$

where $S_{yu}(z)$, called the cross-power spectrum between the y and the u processes, is the z-transform of $R_{yu}(\ell)$ and $S_{uu}(z)$, the z-transform of $R_{uu}(\ell)$, is called the power spectrum of the u process.

With these equations, the estimation of the transfer function $G(z)$ has been converted to the estimation of the correlation functions R_{yu} and R_{uu} or, equivalently, to the estimation of the spectra S_{yu} and S_{uu} from the data records of $y(k)$ and $u(k)$. This topic, spectral estimation, is covered by a vast literature going back to the 1930's at least, and we will indicate only some of the important results relevant to the identification problem here. In the first place, our estimation of the correlation functions depends upon the ergodic assumption that the stochastic expectations can be computed as time averages according to [2]

$$R_{yu}(\ell) = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} y(n)u(n-\ell). \qquad (12.32)$$

In order to use Eq. (12.32) with experimental data, we must consider that the data are finite. Suppose we have N points of input and N points of output data. Then we can write

$$\hat{R}_{yu}(\ell) = \frac{1}{N} \sum_{n=0}^{N} y(n)u(n-\ell), \qquad (12.33)$$

where it is understood that $y(k)$ and $u(k)$ are zero outside the range $0 \leq k \leq N-1$. The number of data values used to compose $R_{yu}(\ell)$ in Eq. (12.33) is $N-\ell$. Clearly, as ℓ gets near to N, this number gets small and very little averaging is involved. We conclude that the estimate of R from Eq. (12.33) can be expected to be very poor for $|\ell|$ near N, and we normally stop the sum for $|\ell| > L/2$, where we might take $L = N/8$, for example. The resulting estimate is a complicated function of the exact number of points used, and typically, in identification, the number to be used is determined by experience and experimentation. Another feature of Eq. (12.33) is the fact that in taking a finite number of points in the sum, it is as if we had multiplied the result, for any given ℓ, by a window of finite duration. The transform reflection of such a product is to convolve the true spectrum (transform) with the transform of the window, and the rectangular window obtained by simply truncating the series is particularly poor for this purpose. Rather, another window is normally introduced that has better spectral smoothing properties, such as the Hamming or the Kaiser window. To illustrate some of the properties of windows, a Hamming window of length 64 and its magnitude spectrum are plotted in Fig. 12.4. It should be recalled that convolution of a function with an impulse returns exactly the same function. Convolution with a bandpass filter of finite width has the effect of averaging the function over the width of the pass band. Fig. 12.4 shows that the Hamming window of length 64 has a spectrum with a main lobe of width about 8/128, or 6.25% of the Nyquist frequency. A longer window would have a more narrow pass band, and a shorter window would have a wider spectral lobe and consequently more averaging of the spectrum in computing the estimate by Eq. (12.33). The engineer must use

2 The MATLAB function xcorr computes $R_{yu}(-\ell) = R_{uy}(\ell)$.

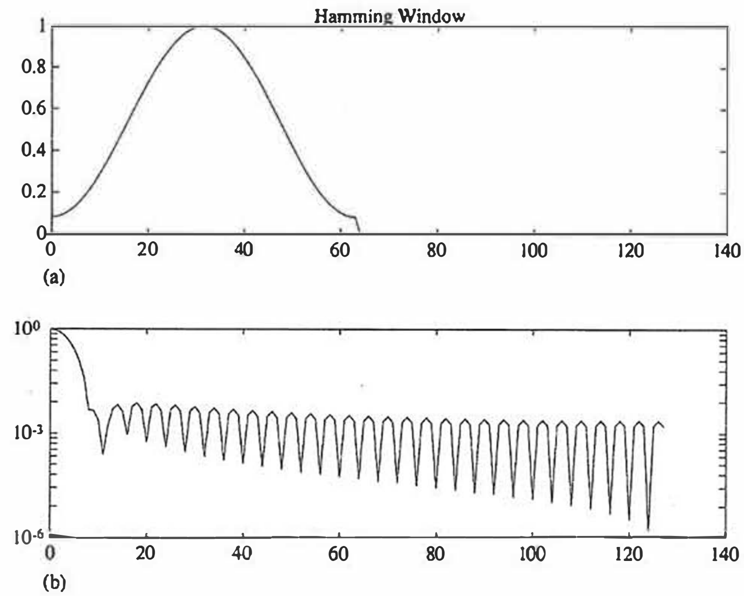


Figure 12.4
Hamming window: (a) weighting function, and (b) spectrum

judgement to select a window function length that will balance noise reduction, for which a short window is wanted, against average spectrum accuracy, for which a long window is wanted.

From these considerations, a practical formula for spectrum estimation is derived as

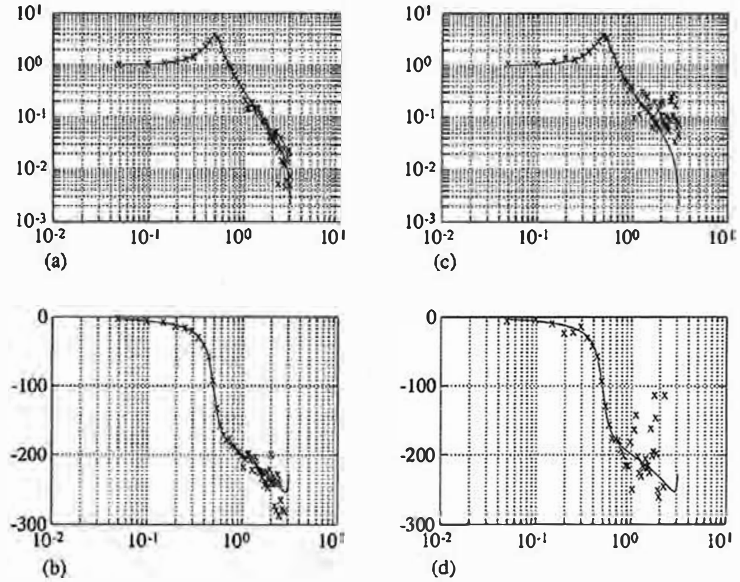
$$\hat{S}_{yu}(z) = \sum_{\ell=-L/2}^{L/2} w(\ell)\hat{R}_{yu}(\ell)z^{-\ell}. \tag{12.34}$$

◆ **Example 12.4** *Stochastic Estimate of a Transfer Function*

Apply a random input signal to the second-order system of Example 12.2 and estimate the transfer function using Eq. (12.31) with the estimates according to Eq. (12.34). Repeat, with 20% random noise added to the output.

Solution. The system was excited with a 256-point random input of zero mean, and the autocorrelation and the cross-correlation were computed for all 512 points for the output with no noise and for the output with 20% noise. That is, the input to the system had a root-mean-square value of one, and the output had a noise added whose root-mean-square value was 0.2. A Hamming window of width 256 was used and the spectra computed for the system

Figure 12.5
Estimation of transfer function with random input using spectral estimation: (a) and (b) no noise, (c) and (d) 20% noise



with no noise and for the system with output noise. The resulting transfer functions computed according to Eq. (12.31) are plotted in Fig. 12.5.

◆

As can be seen from parts (a) and (b) of Fig. 12.5, the estimate when there is no noise is excellent. However, with the noise, as seen in parts (c) and (d), there is serious deviation of the estimate from the true transfer function, especially at the higher frequencies. In a real situation, serious consideration would be given to using a shorter window, perhaps as short as 100 or even less. As the window is shortened, however, the details of the resonant peak, as discussed, will be smeared out. The engineer's judgement and knowledge of the system under investigation must be used to balance the random fluctuations against the systematic errors.

12.3 Models and Criteria for Parametric Identification

Because several design techniques including design by root locus and design by pole placement require a parametric model, it is important to understand how such models can be constructed. This is particularly important for adaptive control where the design must be implemented on-line and either the plant parameters or the controller parameters are to be estimated. To formulate a problem for parametric identification, it is necessary to select the parameters that are to be estimated and a criterion by which they are to be evaluated.

12.3.1 Parameter Selection

For example, suppose we have a transfer function that is the ratio of two polynomials in z. We can select the coefficients of these polynomials as the parameters. For the third-order case

$$G(z, \theta) = \frac{b_1 z^2 + b_2 z + b_3}{z^3 + a_1 z^2 + a_2 z + a_3}, \tag{12.35}$$

and the parameter vector $\boldsymbol{\theta}$ is taken as

$$\boldsymbol{\theta} = (a_1\ a_2\ a_3\ b_1\ b_2\ b_3)^T. \tag{12.36}$$

We imagine that we observe a set of input sample values $[u(k)]$ and a set of corresponding output sample values $[y(k)]$, and that these come from a plant that can be described by the transfer function Eq. (12.35) for some "true" value of the parameters, $\boldsymbol{\theta}^o$. Our task, the parametric identification problem, is to compute from these $[u(k)]$ and $[y(k)]$ an estimate $\hat{\boldsymbol{\theta}}$ that is a "good" approximation to $\boldsymbol{\theta}^o$.[3]

To repeat the formulations from a different point of view, suppose we postulate a (discrete) state-variable description of the plant and take the parameters to be the elements of the matrices $\boldsymbol{\Phi}$, $\boldsymbol{\Gamma}$, $\mathbf{H}$. We assume $J = 0$. Then we have

$$\begin{aligned} \mathbf{x}(k+1) &= \boldsymbol{\Phi}(\boldsymbol{\theta}_1)\mathbf{x}(k) + \boldsymbol{\Gamma}(\boldsymbol{\theta}_1)u(k), \\ y(k) &= \mathbf{H}(\boldsymbol{\theta}_1)\mathbf{x}(k). \end{aligned} \tag{12.37}$$

For the third-order model, there are nine elements in $\boldsymbol{\Phi}$ and three elements each in $\boldsymbol{\Gamma}$ and $\mathbf{H}$, for a total of fifteen parameters in Eq. (12.37), where six were enough in Eq. (12.35) for an equivalent description. If the six-element $\boldsymbol{\theta}$ in Eq. (12.36) and the fifteen-element $\boldsymbol{\theta}_1$ in Eq. (12.37) describe the same transfer function, then we say they are *equivalent* parameters. Any set of $u(k)$ and $y(k)$ generated by one can be generated by the other, so, as far as control of y based on input u is concerned, there is no difference between $\boldsymbol{\theta}$ and $\boldsymbol{\theta}_1$ even though they have very different elements. This means, of course, that the state-variable description has nine parameters that are in some sense redundant and can be chosen rather arbitrarily. In fact, we have already seen in Chapter 4 that the definition of the state is not unique and that if we were to change the state in Eq. (12.37) by the substitution $\boldsymbol{\xi} = \mathbf{T}\mathbf{x}$, we would have the same transfer function $G(z)$. It is exactly the nine elements of $\mathbf{T}$ which represent the excess parameters in Eq. (12.37); we should select these in a way that makes our task as easy as possible. The standard, even obvious, way to do this is to *define* our state so that $\boldsymbol{\Phi}$, $\boldsymbol{\Gamma}$, $\mathbf{H}$ are in

3 For the moment, we assume that we know the number of states required to describe the plant. Estimation of n will be considered later.